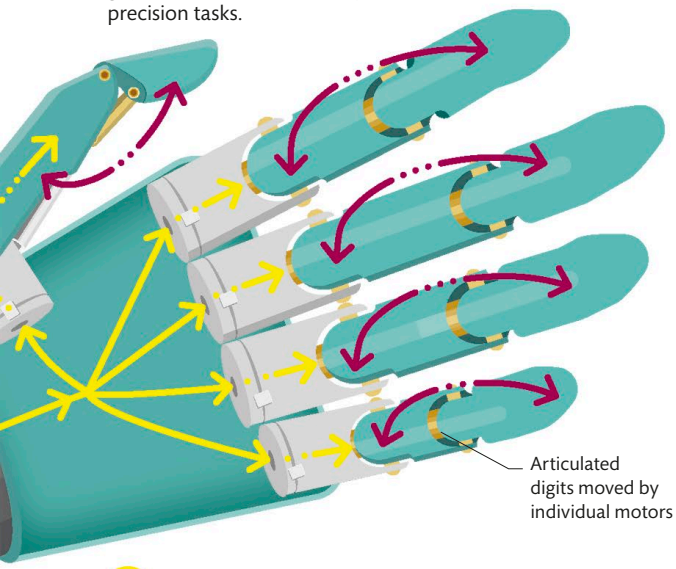
WHEN WERE PROSTHESES FIRST USED?

Artificial body parts were used at least 3,000 years ago. The oldest surviving prosthetic body part is a toe made of wood and leather found on an ancient Egyptian mummy.

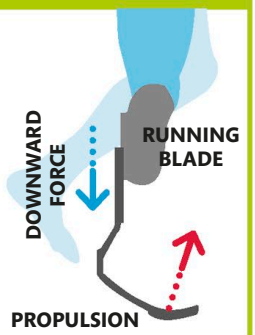
3 Hand movements
The wrist, fingers, and thumb are moved by motors. Some types of prostheses allow the digits to move in unison for powered grips or in a coordinated way for precision tasks.



2 Data sent to microprocessor
The signals from the muscles are sent to the microprocessor, which translates this data into commands to activate motors in the hand and wrist. Different muscle signals can enable different types of grips.

RUNNING BLADES

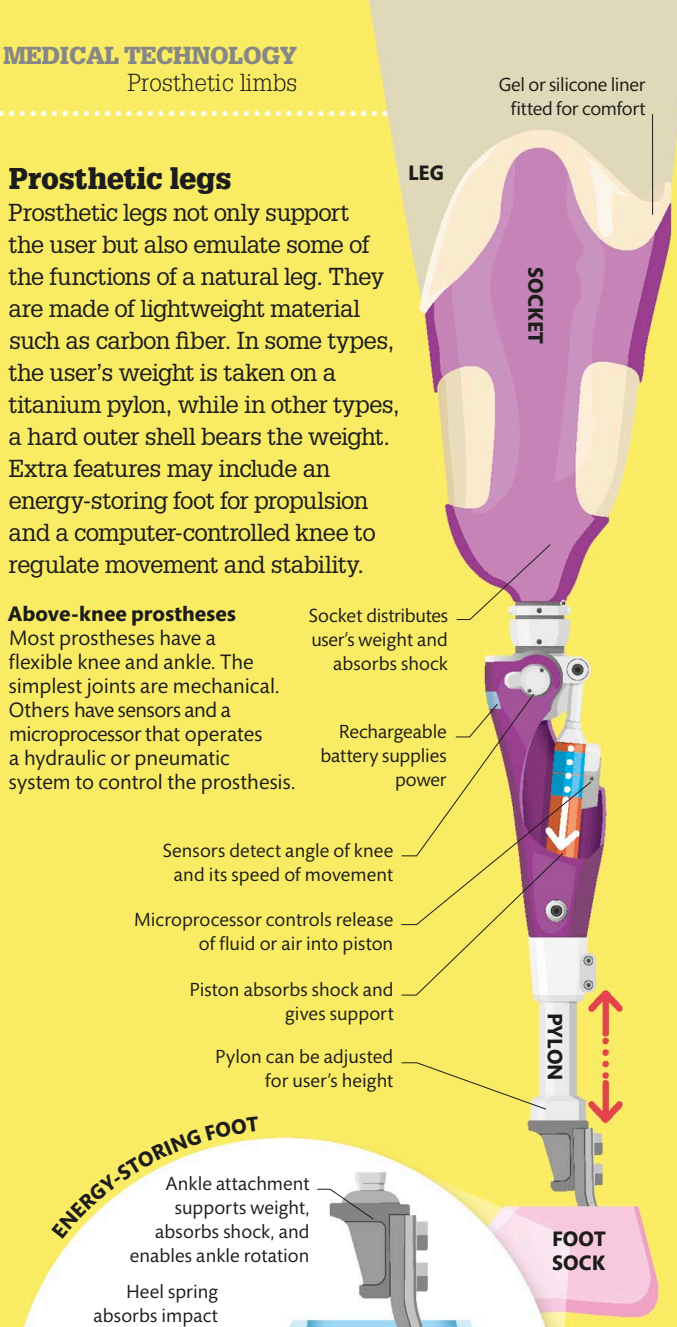
Used by athletes, running blades are made of layers of carbon fiber bonded together, making them light but strong and flexible. The soles have treads or spikes for traction. The blade bends as the runner lands on it, and then as the "foot" rolls, the blade rebounds, releasing energy to power the athlete forward.

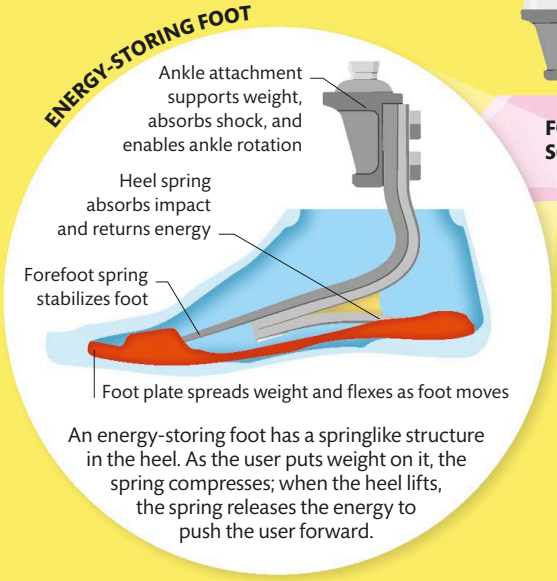


Prosthetic legs

Prosthetic legs not only support the user but also emulate some of the functions of a natural leg. They are made of lightweight material such as carbon fiber. In some types, the user's weight is taken on a titanium pylon, while in other types, a hard outer shell bears the weight. Extra features may include an energy-storing foot for propulsion and a computer-controlled knee to regulate movement and stability.

Above-knee prostheses
Most prostheses have a flexible knee and ankle. The simplest joints are mechanical. Others have sensors and a microprocessor that operates a hydraulic or pneumatic system to control the prosthesis.





An energy-storing foot has a springlike structure in the heel. As the user puts weight on it, the spring compresses; when the heel lifts, the spring releases the energy to push the user forward.